Name the denominator and the averaging law

Testing convention

$f(a, \Theta) = F/\max\{1, R\}$

F: false claims; R: reported claims
Empty report: f = 0

Reported error rate

D/N only if $N > 0$

N = 0: rate undefined
Show disclosure separately

Realized state

$f(a, \Theta)$

One action and
one target state

Posterior

$E[f(a(y), \Theta) \mid Y = y]$

Condition on the
observed data

Joint model

$E_{\Theta, Y}[f(a(Y), \Theta)]$

Average parameters
and data together

Fixed parameter

$E_{Y \mid \theta}[f(a(Y), \theta)]$

Repeat data at
a fixed parameter

Correct conditional law + bound for all data
(or almost surely) → joint-model bound.
No automatic fixed-parameter guarantee.